

Djordje Miladinovic

Curriculum Vitæ

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Summary

- **Profile:** Senior AI/ML research scientist & engineer | PhD in computer science (AI/ML) | Swiss resident | Serbian citizen. 10+ years in AI/ML research | 15+ years in software development | 5+ years in project leading | 5+ years in life sciences.
- **Work:** Designing and leading the end-to-end development of scalable AI/ML and software products, from concept to deployment.
- **Mission:** My goal is to utilize technology, innovative design, and data-driven insights to improve individual and societal well-being, fuel business growth, and deliver best-in-class solutions with measurable impact.

Experience



Senior AI/ML Research Scientist & Engineer, *AI department of GSK plc*

Oct'21-Present

- Spearheaded the development of a foundational model integrating omics data to accelerate preclinical drug discovery, enabling target identification and uncovering disease and drug mechanisms – *adopted by three R&D units at GSK, enhancing decision-making and aiding in the identification of optimal candidate protein targets.*
- Supported clinical efforts to optimize the use of existing GSK drugs by applying survival analysis and processing large-scale electronic health record (EHR) databases – *to be used by clinicians for improved diagnosis and prognosis.*



Machine Learning Researcher, *ETH Zürich*

Feb'17-Jun'21

- See the full list of my publications at [Google Scholar](https://scholar.google.com/citations?user=...). Notable projects include:
- **Sleep-learning web platform:** Led a team of researchers to develop a web-based large-scale machine learning platform for sleep pattern recognition from EEG/EMG signals – *now commercialized with over 100K jobs processed.*
- **SDN neural networks:** Co-developed a neural architecture for realistic image synthesis – *state-of-the-art performance.*
- **Granger-causality framework for multimodal analysis of mass spectrometry and EEG data:** Co-developed an explainable AI (XAI) algorithm to elucidate causal relationships between metabolism and sleep.



Machine Learning Researcher (Visitor), *Max Planck Institute for Intelligent Systems* **Sep'18-Dec'18**

- Co-developed a [framework for evaluating disentangled representations](#) – *has about 200 scientific citations so far.*



Machine Learning Researcher & Engineer (Intern), *Logitech Europe S.A.*

Sep'16-Feb'17

- Developed an AI/ML product to predict potential premium users and reduce churn rate.



Machine Learning Researcher (Visitor), *Disney Research Studios, Zurich*

Jan'16-Sep'16

- Developed a software platform for detecting mechanical failures in Disney's humanoid robots based on IMU sensor readings. The platform uses novel machine learning algorithm to assess robot degradation in a human-like way.

Education



PhD in Computer Science – Machine Learning, *ETH Zürich*

Sep'17-Jun'21

- **Thesis:** "On training Deep Generative Models with Latent Variables" – [Prof. Joachim M. Buhmann](#).
- **Research areas:** Generative image and text modeling | variational autoencoders | representation learning | deep learning | language models | machine learning for biology | drug discovery;



MS in Computer Science – Machine Learning, *ETH Zürich*

Sep'13-Jun'16

- **Thesis:** "Perceptual Analysis Framework for Discovering Anomalies in Humanoid Motions" – [Prof. Otmar Hilliges](#).
- **Focus:** Machine learning | software engineering | operating systems | distributed computing;



BS in Computer Science – Software Engineering, *ETF Belgrade*

Sep'09-Sep'13

- **Thesis:** "Java-based Interactive Software Simulator of a CISC Processor" – [Prof. Zaharije Radivojević](#).
- **Focus:** Software engineering and design | algorithms | data structures | operating systems | hardware design;

Technical Skills

Programming	<i>Advanced:</i> Python (PyTorch); <i>Good but rusty:</i> C/C++ Java Javascript Bash SQL;
Infrastructure	<i>Data formats:</i> Parquet H5 H5AD CSV YAML JSON Pickle;
(Varying levels)	<i>Databases:</i> PostgreSQL MongoDB SQLite Milvus; <i>Big data:</i> Spark Hive Hadoop; <i>AI/ML frameworks:</i> Pytorch Lightning Ray Weights & Biases MLflow HuggingFace Kubeflow; <i>High Performance Computing (HPC):</i> SLURM LSF Condor; <i>Middleware:</i> Celery Redis Kafka; <i>Cloud:</i> GCP/GCS AWS/S3 Microsoft Azure; <i>Deployment:</i> Docker Kubernetes; <i>CI/CD:</i> GitHub Actions CircleCI; <i>Web:</i> Node.js Django HTML/CSS;
Python libraries	<i>Data Science:</i> Pandas SciPy Seaborn NumPy Scanpy Scrapy OpenCV; <i>AI/ML:</i> PyTorch Scikit-learn SpaCy Fairseq LangChain Tensorflow;
Design tools	Figma Inkscape Blender PowerPoint Adobe Illustrator;

Languages

Serbian (Native) | English (Fluent) | German (Intermediate) | Spanish (Beginner);

Hobbies

Waterpolo | Swimming | Basketball | Tennis | Chess | Cinematography | Reading;